

Meeting Summaries of Jan 8, 2026

Date: Jan 8, 2026

Time: 14:30–15:15 PM (PDT)

Attendees: Prof. Jan, Dr. Javed and Feng

Topic: Feng gave an update on the work done in winter break and showed the 90% framework of the thesis work with the resulted figures and tables.

Summary of Discussion

- Feng presented the general results that covered the previous analysis reported in meetings in November and December. It consolidated the **4** major stages:
 1. **01 Contamination Assessment:** Define and compute the computation of pollution scores for each site, preserve the orthogonal pollution profiles (pollution PCs).
 2. **02 Taxa Assemblage in Reference Sites:** Select least polluted sites and cluster the sites based on their taxa assemblage, [set](#) and interpret the natural taxa assemblage groups.
 3. **03 Environmental Drivers of Taxa Assemblage:** Reveal the habitat features that significantly drive species composition (RDA) and classify the sites into selected clusters.
 4. **04 Community Composition Measures:** [In each cluster](#), decompose the species matrix and sum the decomposed signals (species PCs) to construct zoobenthic community index (ZCI) for each site.
 5. **05 Regression between ZCI* and Pollution Scores* (ongoing):** [In each cluster](#), train regression models (piece-wise quantile regression) between ZCI* and Pollution Scores*. The current focus is to apply regressions in 3 levels of scope:
 - Comprehensive ZCI vs Comprehensive Pollution Scores ($\mathcal{R}^1 \rightarrow \mathcal{R}^1$): More loss in information but clarity in model building.
 - Individual ZCI(p) vs Comprehensive Pollution Scores ($\mathcal{R}^1 \rightarrow \mathcal{R}^p$): More information preserved but harder to build models.
 - Individual ZCI vs Individual Pollution Scores(q) ($\mathcal{R}^q \rightarrow \mathcal{R}^p$): Most information preserved but hardest to build models.
- Prof. Javed suggested to be careful with using AI in writings. The content should always be self-generated first, then AI can be used to polish the language, not to generate the content directly.
- Prof. Jan advised to make the figures and fontsize larger to let readers see clearly. The analytical framework and results should be consistent with the ecological facts that are known. It requires Feng to check the directly related literature and findings to support the interpretations and validate the model settings and interpretations of the results.

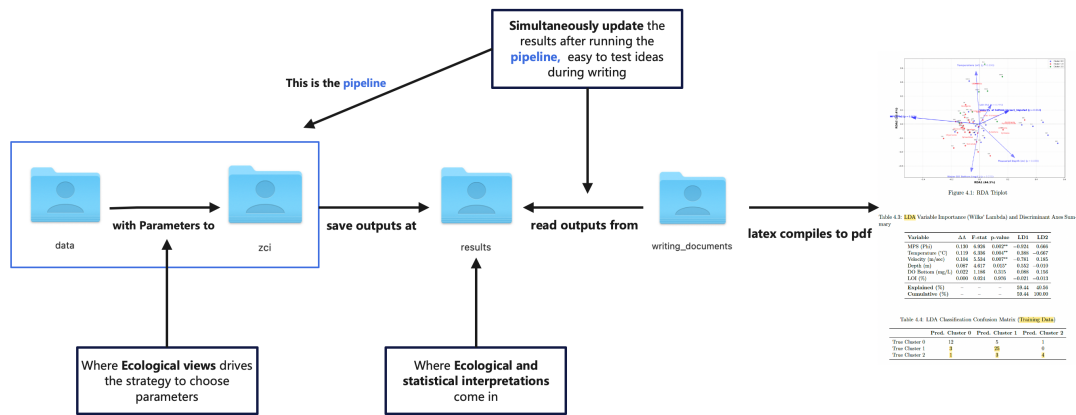


Figure 1: Overview of the system-level code pipeline supporting an end-to-end workflow from data and parameters to documentation.

- Prof. Jan discussed in more detail the selection of reference sites and its importance to following results. To safely select the least-polluted sites as reference sites, **only 20% or less** of sites should be selected to study the natural taxa assemblage (involving stages 02 and 03).

When the overall sample size is small (e.g., less than 100), choosing 20% might lead to **insufficient** number of sites in subsequent analyses, raising the question: *How many sites are enough* to stably estimate the results we could get if we had more sites (e.g., 200 or more)?

This **sample-size question** is **the core issue** that runs through the entire study—not only in the hypothesis-power tests of regression models, but it implicitly determines the final properties and performance of ZCI vs Pollution Scores. This implicit determination may propagate from the selection of reference sites, to the selection of clusters, and finally to the regression models.

Action Items

- Feng needs to make the figures and their elements larger in drafts so readers can see clearly.
- Feng should cross-check existing results ('facts') in related literature to validate and optimize analysis parameters and framework.
- Feng needs to understand the ecological roles of the species and know how to orally interpret and defend results about these species.
- The regression module linking ZCI* and Pollution Scores* should be completed and compiled into the framework as soon as possible.

Follow-Up

Temporary meeting time: Jan 15, 14:30–15:00 PM (PDT)

Supportive Materials

- A meeting recording with subtitles has been stored locally.